The examination is being carried out on the **following application documents**

Description, Pages

1-19 as published

Claims, Numbers

1-7 as published

Drawings, Sheets

1/7-7/7 as published

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 2011/132954 A1 (TOMEI MAOKO [JP] ET AL) 9 June 2011 (2011-06-09)

D2 US 2012/312141 A1 (TOMEI NAOKO [JP]) 13 December 2012 (2012-12-13)

D3 Wilks E M ET AL: "The resistance of diamond to abrasion", J. Phys. D: Appl. Phys., 26 April 1972 (1972-04-26), pages 1902-1928, XP093237919,

D4 EP 3 015 236 A1 (MITSUBOSHI DIAMOND IND CO LTD [JP]) 4 May 2016 (2016-05-04)

D5 US 2015/274535 A1 (SATO TAKESHI [JP] ET AL) 1 October 2015 (2015-10-01)

INDEPENDENT CLAIMS

CLAIM 1

1 *Novelty (Art. 54(2) EPC)*

The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 is not new within the meaning of Article 54(1) and (2) EPC.

1.1 **D1** (par. 21, par. 52-par. 54, *par. 56-57, Figs. 1-3*) discloses:

A contactor having an annular shape *(par. 52, Fig. 1, Fig. 2)* and made of a polycrystalline diamond including a plurality of diamond particles *(par. 21)*, the contactor being configured to have a rotation axis extending through a center of the contactor (see Fig. 5), the contactor comprising:

- a first portion including an inner end portion, the first portion having a constant thickness in a radial direction, *(see Fig. 1, Fig. 2)*; and
- a second portion including an outer end portion, the second portion having a decreasing thickness in the radial direction *(see Fig. 2)*, wherein the second portion has
- a first surface continuous with an upper surface of the first portion,
    - a second surface continuous with a lower surface of the first portion, and
    - a connection surface connecting the first surface and the second surface to each other and including the outer end portion,
      wherein,
- in a cross-section along the rotation axis,
    - an angle θ formed by a first line segment indicating the first surface and a second line segment indicating the second surface is 100° to 150° *(par. 53 "The angle θ of the edge of this blade 11 is usually obtuse, for example within a range of 90° to 160°")*,
    - a length between a first boundary portion and a second boundary portion is 1 µm to 10 µm, the first boundary portion being a boundary between the first surface and the connection surface, the second boundary portion being a boundary between the second surface and the connection surface,

(par. 56 and Fig. 1 and 3 disclose different types of grooves 13 created in the blade edge 12 of the scribing wheel. Par. 57 discloses that the depth D of the grooves is 25 µm or more. Thus, a length between a first boundary portion and a second boundary portion differs in D1 along the circumference of the wheel edge, increasing from a minimal value at the blade edge 12 toward the full depth D of the groove 13 (25 µm or more). Thus, at certain position along the circumference of the wheel edge, there implicitly has to be a length between a first boundary portion and a second boundary portion between 1 µm to 10 µm.)

- and a length from the rotation axis to the outer end portion is 0.5 mm to 5 mm

(par. 54 "outer diameter of the scribing wheel 1 is 1 mm to 5 mm." It is acknowledged that a length from the rotation axis to the outer end portion in D1 is not constant, it changes along the circumference of the wheel, however this feature is not disclosed in the claim), and wherein

- the plurality of diamond particles have an average particle diameter of 10 nm to 300 nm *(par. 21 "the average particle diameter of the diamond grains is 0.5 µm or less").*

1.1.1 Claim 1 is not novel over D1.

1.2 **D2** (par. 37-38, par. 43, par. 49, *Fig. 1B, Fig. 2B*) also discloses all features of claim 1:

A contactor having an annular shape *("wheel")* and made of a polycrystalline diamond including a plurality of diamond particles *(par. 37, It is to be noted that par. 27 of the application discloses that "The polycrystalline diamond may include a plurality of diamond particles and a binder. Thus, the therm "made of" is interpreted as "comprising" rather than "consisting")*, the contactor being configured to have a rotation axis extending through a center of the contactor *(central axis 12a, see Fig. 1B),* the contactor comprising:

- a first portion including an inner end portion, the first portion having a constant thickness in a radial direction, *(see Fig. 1B)*; and
- a second portion including an outer end portion (see Fig. 1B), the second portion having a decreasing thickness in the radial direction *(see Fig. 1B)*, wherein the second portion has
  - a first surface continuous with an upper surface of the first portion *(one side of V-shaped edge)*,
  - a second surface continuous with a lower surface of the first portion *(the other side of V-shaped edge)*, and
  - a connection surface connecting the first surface and the second surface to each other and including the outer end portion *(see Fig. 2B showing the enlarged sectional view of a ridge part of a cutting edge after grinding process),*
    wherein,
- in a cross-section along the rotation axis,
  - an angle θ formed by a first line segment indicating the first surface and a second line segment indicating the second surface is 100° to 150° *(par. 38: 85-150 degrees)*,

- a length between a first boundary portion and a second boundary portion is 1 µm to 10 µm, the first boundary portion being a boundary between the first surface and the connection surface, the second boundary portion being a boundary between the second surface and the connection surface *(par. 38: width w1 is within a range from 10 µm to 30 µm and in another embodiment of par. 43 is a width w4 within a range from 2 µm to 10 µm),*

- and a length from the rotation axis to the outer end portion is 0.5 mm to 5 mm *(par. 49: an outer diameter of 2.7 mm)*, and wherein
- the plurality of diamond particles have an average particle diameter of 10 nm to 300 nm *(par. 37: core diamond particles having a particle size within a submicron range).*

1.2.1 Claim 1 is also not novel over D2.

1.3 **D3** (*page 1903: second paragraph of Section "Experimental method; second paragraph on page 1904; Fig. 1*) also discloses all features of claim 1:

A contactor having an annular shape *("wheel")* and made of a polycrystalline diamond including a plurality of diamond particles *(page 1903: second paragraph of Section "Experimental method. It is to be noted that par. 27 of the application discloses that "The polycrystalline diamond may include a plurality of diamond particles and a binder. Thus, the therm "made of" is interpreted as "comprising" rather than "consisting")*, the contactor being configured to have a rotation axis extending through a center of the contactor *(page 1903: second paragraph of Section "Experimental method),* the contactor comprising:

- a first portion including an inner end portion, the first portion having a constant thickness in a radial direction, *(page 1903: second paragraph of Section "Experimental method)*; and
- a second portion including an outer end portion *(V-shaped edge, page 1903: second paragraph of Section "Experimental method)*, the second portion having a decreasing thickness in the radial direction *(V-shaped edge)*, wherein the second portion has
- a first surface continuous with an upper surface of the first portion *(one side of V-shaped edge)*,
    - a second surface continuous with a lower surface of the first portion *(the other side of V-shaped edge)*, and

- a connection surface connecting the first surface and the second surface to each other and including the outer end portion *(D2 discloses in second paragraph on page 1904 that after only a few cuts the edge becomes blunt, thus forming the connection surface, see also Fig. 1.),*
  wherein,

- in a cross-section along the rotation axis,
  - an angle θ formed by a first line segment indicating the first surface and a second line segment indicating the second surface is 100° to 150° *(angle of about 110°, page 1903: second paragraph of Section "Experimental method)*,
  - a length between a first boundary portion and a second boundary portion is 1 µm to 10 µm, the first boundary portion being a boundary between the first surface and the connection surface, the second boundary portion being a boundary between the second surface and the connection surface,

(D3 discloses in second paragraph on page 1904 that after only a few cuts the edge becomes blunt, see also Fig. 1. Thus, the thickness of the blade gets into the given range when the originally sharp edge gets blunted)

- and a length from the rotation axis to the outer end portion is 0.5 mm to 5 mm *(page 1903: second paragraph of Section "Experimental method "small wheel of about 1 cm radius")*, and wherein
- the plurality of diamond particles have an average particle diameter of 10 nm to 300 nm *(page 1903: second paragraph of Section "Experimental method).*

1.3.1 Claim 1 is also not novel over D3.

1.4 Document **D4** (par. 21-26, Fig. 1-3) discloses all features of claim 1, except explicitly disclosing the average particle diameter of diamonds particles. However, D4 discloses the wheel made of polycrystalline diamond (par. 22), wherein it is generally known that the polycrystalline diamond has average diamond grains between 80 nm to 1 µm (see e.g. par. 54 of D5), thus also within the claimed ranges 10 nm to 300 nm.

1.4.1 Claim 1 is thus not novel also over D4.

CLAIM 5

2 *Novelty (54(2) EPC)*

2.1 **D3** *(page 1903: second paragraph of Section "Experimental method, page 1903: third paragraph of Section "Experimental method)* discloses:

An evaluation method of an abrasion characteristic of a monocrystalline diamond *(page 1903: second paragraph of Section "Experimental method)* in a minute region *(the dimensions of the wheel and the cut are small which implies testing in a minute region)*, the evaluation method comprising:
a first step of forming an abrasion mark on the monocrystalline diamond by, while rotating the contactor according to any one of claim 1 to claim 4 (*see reasoning for claim 1 above)*,

- pressing the monocrystalline diamond against the outer end portion of the contactor *("The diamond under test is mounted on a balance arm which is brought against the abrading wheel under a known load of the order of 100 g": page 1903: third paragraph of Section "Experimental method)*; and
- a second step of evaluating the abrasion characteristic of the monocrystalline diamond in the minute region based on a length of the abrasion mark (*page 1903: third paragraph of Section "Experimental method)*.

2.2 The subject-matter of claim 5 is therefore not novel.

3 DEPENDENT CLAIMS

The following dependent claims do not contain any additional features, which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to novelty and/or inventive step.

3.1 **Claim 2:** the given Knoop hardness values relates to usual range of the polycrystalline diamond, see D5 (Table 1)

3.2 **Claim 3:** D1 (see Fig. 3a-c) and D3 (blunted edge) discloses a straight line of the connecting surface. It is to be noted that claim 3 is unclear, because an intersection (of the connection surface and a cross section) relates to a single point, not to a line. Further it is unclear to what object and to which axis the cross-section relates.

3.3 **Claim 4** (the length from the rotation axis to the outer end portion is larger than a length from the rotation axis to the first boundary portion and larger than a length from the rotation axis to the second boundary portion): D1 (see also the grooves in the layout of Fig. 1), D2 (see Fig. 3B)), D3 (V-shaped edge wheel of D3 discloses these features) and D4 (see Fig. 2)

3.4 **Claim 6**: see D3: page 1903: third paragraph of Section "Experimental method

3.5 **Claim 7:** D3: see Table 9

4 GENERAL

4.1 It is not at present apparent which part of the application could serve as a basis for a new, allowable independent claim. Should the applicant nevertheless regard some particular matter as patentable, new claims could be filed. The applicant should indicate how the subject-matter of the new claim differs from the state of the art and the significance thereof. **Arguments as to why the independent claims are inventive should be provided by using the problem-and-solution approach (GL G-VII, 5).**

4.2 In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant should clearly **identify the amendments carried out**, irrespective of whether they concern amendments by addition, replacement or deletion, and to **indicate the passages** of the application as filed on which these amendments are based.

4.3 The applicant is also invited to **modify the description in order to take into account the amended set of claims**, as required by Rule 42(1(c)) EPC, in order to accelerate the procedure.

4.4 The independent claims should be drafted in the two-part form in accordance with Rule 43(1) EPC, with those features or method steps known in combination from the prior art begin placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterising part (Rule 43(1)(b) EPC).